DO NOW — RECALL

Programming Techniques · Sequence and Selection

Name: _____ Class: _____ Date: _____

1. What does a full adder include that a half adder does not?

2. What is a local variable?

3. Why does a stack use LIFO?

4. Simplify A ∨ (A ∧ B).

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Carry-in

2. Model answer

A variable declared within a scope

3. Model answer

The most recently inserted item is removed first

4. Model answer

A, by absorption

By the end, you can…

• Trace instructions in sequence

• Choose IF/ELSE conditions

• Write a short C# decision algorithm

Vocabulary — know these terms

Recall the term — sequence

Recall the term — selection

Recall the term — Boolean

Recall the term — condition

Recall the term — trace

NEW LEARNING

Read, annotate and explain

Follow the order of execution

int score = 7; score = score + 3; if (score >= 10) Console.WriteLine("Pass");

Explain the important decision

Sequence changes score before selection tests it. The condition is true at exactly 10 because >= includes the boundary.

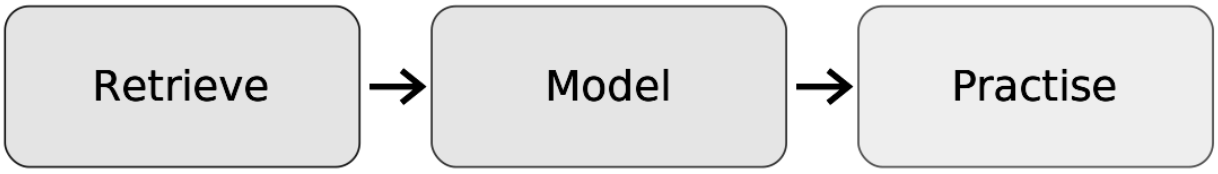
Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Follow the order of execution

int score = 7; score = score + 3; if (score >= 10) Console.WriteLine("Pass");



Three-stage diagram showing retrieve, model, and practise

I DO — WORKED EXAMPLE

Follow the worked method

Follow the order of execution

Update in sequence

score starts at 7. Adding 3 changes the same variable to 10; the condition has not been tested yet.

Evaluate the condition

Substitute the new value: 10 >= 10 is true. Using the old value 7 would trace the wrong state.

Take the selected branch

The true branch prints Pass. For initial scores 6, 7 and 8, the updated scores are 9, 10 and 11: only the latter two print Pass.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

1. Problem 1

value = 3; value *= 4; is value > 10?

Method / working:

2. Problem 2

A user enters age 16. Test age >= 18.

Method / working:

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Problem 1 value = 3; value *= 4; is value > 10? Discuss, then respond

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

Guided 2: Problem 2 A user enters age 16. Test age >= 18.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Apply independently
Trace x = 4; x = x + 7; output x. [2]

Question 2

Apply independently
Write a C# condition that prints Adult when age >= 18. [3]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Apply independently
Explain one difference between sequence and selection. [2]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

True or false: an else branch runs when its if condition is false.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Check your thinking

Yes: 12 > 10.

IMPROVE: Explain one correction to your own reasoning.

Guided 2 — original CHECK

Discuss, then respond

A user enters age 16. Test age >= 18.

Check your thinking

False; else branch executes.

IMPROVE: Explain one correction to your own reasoning.